

Database Systems - MCQ Question Bank (Set 2)

20 Multiple Choice Questions | Includes 3 repeated from Set 1 | Answer Key on Last Page

1. Which key uniquely identifies each record in a table and cannot contain NULL values?

- A. Foreign Key
- B. Candidate Key
- C. Primary Key
- D. Alternate Key

2. Which of the following is NOT one of the ACID properties of a transaction?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Availability

3. Which indexing structure is most commonly used by databases for efficient range queries?

- A. Hash Index
- B. B+ Tree Index
- C. Bitmap Index
- D. Linked List Index

4. Which of the following is used to remove duplicate rows from a SQL query result?

- A. UNIQUE
- B. DISTINCT
- C. DIFFERENT
- D. FILTER

5. A schedule is said to be serializable if it is equivalent to some:

- A. Concurrent schedule
- B. Serial schedule
- C. Partial schedule
- D. Deadlocked schedule

6. Which of the following best describes a composite key?

- A. A key made up of two or more attributes that together uniquely identify a row
- B. A key that references another table
- C. A key generated automatically by the system
- D. A key with only numeric values

7. In BCNF (Boyce-Codd Normal Form), every determinant must be:

- A. A foreign key
- B. A candidate key
- C. A composite attribute
- D. A non-key attribute

8. Which of the following SQL statements is used to modify existing records in a table?

- A. MODIFY

- B. ALTER
- C. UPDATE
- D. CHANGE

9. What does the term 'referential integrity' ensure in a relational database?

- A. All tables have the same number of columns
- B. Foreign key values match a primary key value in the referenced table, or are NULL
- C. Data is encrypted during transmission
- D. Queries execute within a fixed time limit

10. Which of the following is an example of a DML (Data Manipulation Language) command?

- A. CREATE
- B. GRANT
- C. INSERT
- D. DROP

11. A checkpoint in database recovery is primarily used to:

- A. Encrypt the transaction log
- B. Reduce the time needed for recovery after a crash
- C. Speed up query optimization
- D. Create a new user account

12. Which type of database anomaly occurs when deleting a record unintentionally removes other useful data?

- A. Insertion anomaly
- B. Update anomaly
- C. Deletion anomaly
- D. Redundancy anomaly

13. Which SQL aggregate function returns the number of rows matching a condition?

- A. SUM()
- B. COUNT()
- C. AVG()
- D. TOTAL()

14. A weak entity in an ER diagram is one that:

- A. Has its own primary key
- B. Cannot be uniquely identified without a related strong entity
- C. Has no attributes
- D. Cannot participate in any relationship

15. Which isolation level allows a transaction to read uncommitted changes made by another transaction?

- A. Serializable
- B. Repeatable Read
- C. Read Committed
- D. Read Uncommitted

16. What is the main purpose of the SQL GROUP BY clause?

- A. To sort query results

- B. To combine rows sharing common values into groups for aggregation
- C. To filter rows before aggregation
- D. To join two or more tables

17. Which of the following storage structures maps search keys directly to record locations using a hash function?

- A. B+ Tree
- B. Hash Index
- C. Heap File
- D. Sequential File

18. In database systems, sharding refers to:

- A. Encrypting a database
- B. Partitioning a database horizontally across multiple servers
- C. Creating a backup copy of the database
- D. Merging two databases into one

19. Which of the following best describes a stored procedure?

- A. A precompiled set of SQL statements stored in the database and executed on demand
- B. A temporary table used in a query
- C. A type of index used for fast lookups
- D. A rule enforcing data type constraints

20. Which of the following is a characteristic of a distributed database?

- A. Data is stored on a single centralized server
- B. Data is stored across multiple physical locations but appears as one logical database
- C. It cannot support concurrent transactions
- D. It does not support SQL queries

Answer Key

1. C 2. D 3. B 4. B

5. B 6. A 7. B 8. C

9. B 10. C 11. B 12. C

13. B 14. B 15. D 16. B

17. B 18. B 19. A 20. B